



< Previous









Next >

5.2.1 Gaussian elimination

 Bookmark this page

Pursue a verified certificate

Verified Certificate

THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

**Maurilio Pugliesi**
Professor
UniversityX

**Justine Doe, Ph.D.**
Professor
UniversityX

Verified Certificate

Valid Certificate ID

12345678901234567890

→ Unlimited access to all course materials

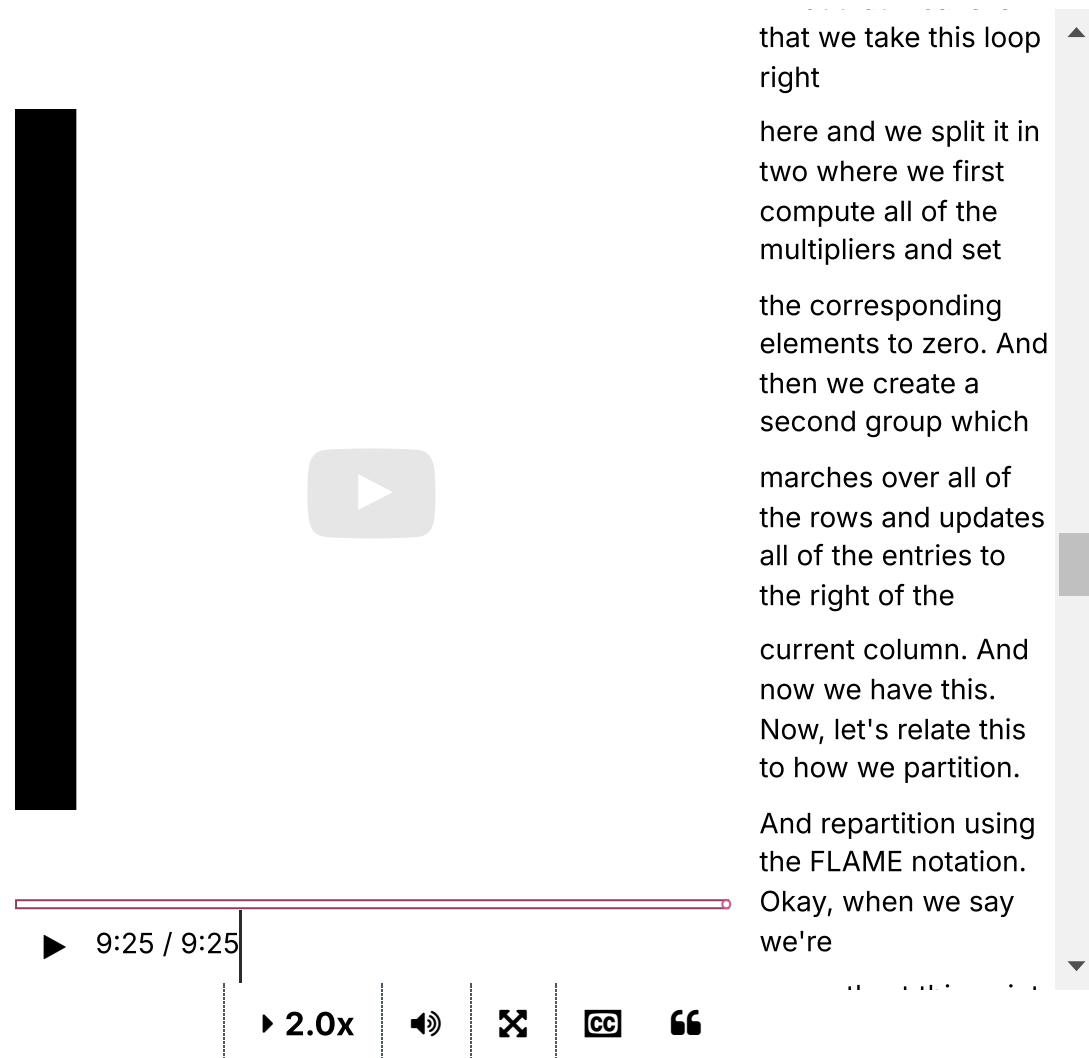
→ Shareable [verified certificate](#)

→ AI learning assistant and translations

Upgrade for \$99

Week 05 due Feb 12, 2025 20:19 IST

Video



Video

 [Download video file](#)

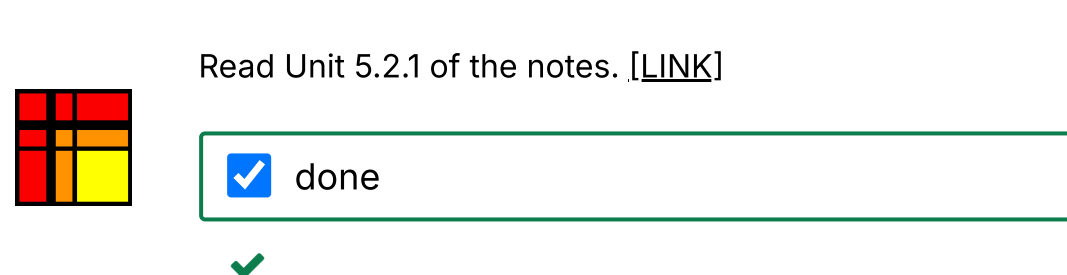
Transcripts

 [Download SubRip \(.srt\) file](#)

 [Download Text \(.txt\) file](#)

Reading assignment

1/1 point (graded)



Submit

✓ Correct (1/1 point)

Discussion

Topic: Week 05 / 05.2.1

Hide Discussion

Add a Post

Show all posts

by recent activity ▼

There are no posts in this topic yet.



Homework 5.2.1.1

1/1 point (graded)
Solve

$$\begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \end{pmatrix} = \begin{pmatrix} -6 \\ 2 \\ 0 \end{pmatrix}.$$

$\chi_0 =$ ✔ Answer: -1

$\chi_1 =$ ✔ Answer: 2

$\chi_2 =$ ✔ Answer: -2

Solution:

For a complete solution, see [the notest](#).

Submit

i Answers are displayed within the problem

Homework 5.2.1.2

1/1 point (graded)
Apply the algorithm

$A = GE(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$
 A_{TL} is 0×0
while $n(A_{TL}) < n(A)$
 $\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$
 $a_{21} := l_{21} = a_{21}/\alpha_{11}$
 $A_{22} := A_{22} - a_{21}a_{12}^T$
 $\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$
endwhile

to the matrix

$$\begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix}$$

and report the resulting matrix.

2	-1	1
✔ Answer: 2	✔ Answer: -1	✔ Ar
-2	-2	3
✔ Answer: -2	✔ Answer: -2	✔ Ar

2	-1	-1
✓ Answer: 2	✓ Answer: -1	✓ Ar

Compare the result to the upper triangular matrix computed in the solution of Homework 5.2.1.1.

Solution:

For a complete solution, see [the notes](#).

Submit

Answers are displayed within the problem

Homework 5.2.1.3

1/1 point (graded)
Applying the algorithm in the last Homework to the matrix

$$A = \begin{pmatrix} 2 & -1 & 1 \\ -4 & 0 & 1 \\ 4 & 0 & -2 \end{pmatrix}$$

yielded

$$\begin{pmatrix} 2 & -1 & 1 \\ -2 & -2 & 3 \\ 2 & -1 & -1 \end{pmatrix}.$$

This can be thought of as an array that stores the unit lower triangular matrix L below the diagonal (with implicit ones on its diagonal) and upper triangular matrix U on and above its diagonal:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 2 & -1 & 1 \\ 0 & -2 & 3 \\ 0 & 0 & -1 \end{pmatrix}$$

Compute $B = LU$ and compare it to A .

$B =$

2	-1	1
✓ Answer: 2	✓ Answer: -1	✓ Ar
-4	0	1
✓ Answer: -4	✓ Answer: 0	✓ Ar
4	0	-2
✓ Answer: 4	✓ Answer: 0	✓ Ar

Solution:

For a complete solution, see [the notes](#).

Submit

Answers are displayed within the problem

[< Previous](#)

[Next >](#)

© All Rights Reserved



edX

[About](#)

[Affiliates](#)

[edX for Business](#)

[Open edX](#)

[Careers](#)

[News](#)

Legal

[Terms of Service & Honor Code](#)

[Privacy Policy](#)

[Accessibility Policy](#)

[Trademark Policy](#)

[Sitemap](#)

[Cookie Policy](#)

[Your Privacy Choices](#)

Connect

[Idea Hub](#)

[Contact Us](#)

[Help Center](#)

[Security](#)

[Media Kit](#)



© 2024 edX LLC. All rights reserved.

深圳市恒宇博科技有限公司 [粤ICP备17044299号-2](#)